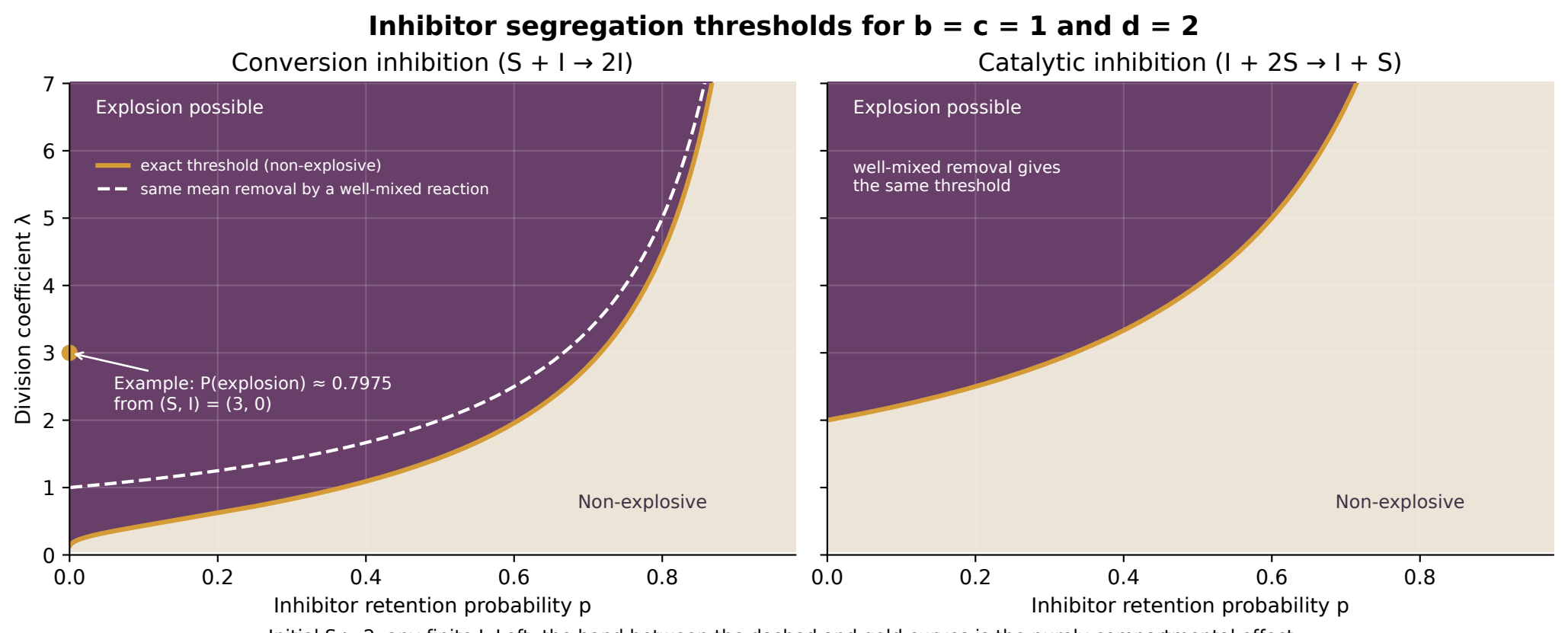
Inhibitor segregation thresholds for b = c = 1 and d = 2
Conversion inhibition (S + I → 2I)
Catalytic inhibition (I + 2S → I + S)
Explosion possible
exact threshold (non-explosive)
same mean removal by a well-mixed reaction
Example: P(explosion) ≈ 0.7975
from (S, I) = (3, 0)
Non-explosive
Explosion possible
well-mixed removal gives
the same threshold
Non-explosive
Division coefficient λ
Inhibitor retention probability p
Inhibitor retention probability p
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
Initial S ≥ 2, any finite I. Left: the band between the dashed and gold curves is the purely compartmental effect.